

Hybrid FXC-WXC Networks with Selective Wavelength Conversion: Dynamic Path Control and Experimental Demonstration

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Abstract We propose a hybrid FXC-WXC architecture with selective wavelength conversion for low-loss and flexible optical networks. Network simulations show reduced blocking and improved OSNR under dynamic control, while transmission experiments demonstrate lower BER and longer reach than WXC-only networks, supporting ZR+ deployment in metro/core networks. ©2026 The Authors

Introduction

Cost-efficient pluggable coherent transceivers such as ZR/ZR+ have attracted significant attention not only for datacentre interconnection but also for metro and core optical networks [1-6]. However, compared with conventional transponders for long-haul transmission, they have a smaller transmission margin [5,6]. Consequently, conventional wavelength cross-connect (WXC)-based networks, despite their high wavelength-granular switching flexibility, are not always favourable because of their large loss [7,8].

To address this issue, we previously proposed a hybrid FXC-WXC network architecture in which most nodes are configured as low-loss fibre cross-connects (FXCs), while lossy WXCs are deployed only at a limited number of nodes [9,10]. Under static optical path design, this architecture achieves both high resource-utilisation efficiency and high signal quality by combining the routing flexibility of WXCs with the low-loss property of FXCs. Under dynamic control, however, wavelength contention becomes more likely over time [11-13], and is more difficult to resolve because the coarse-granularity switching of FXCs limits routing flexibility [14]. Increasing the number of WXC nodes can mitigate this problem, but at the expense of higher insertion loss, cost, and power consumption. By contrast, wavelength conversion locally relaxes the wavelength continuity constraint and resolves contention where it occurs [15-20]. Regenerative wavelength conversion can also restore signal quality and extend the feasible transmission reach [21,22].

In this paper, we propose a hybrid FXC-WXC network with a limited number of WXC nodes and selective wavelength-conversion resources. Unlike our previous work on static network design, this paper investigates dynamic optical path control and introduces selective wavelength conversion at WXC nodes to mitigate wavelength contention while preserving the low-loss property of the hybrid FXC-WXC architecture. Numerical simulations show that selective wavelength conversion suppresses the reduction in the number

of accommodated optical paths from 13.7% to 5.2% while expanding the optical signal-to-noise ratio (OSNR) margin. Transmission experiments further demonstrate improved transmission performance compared with that of a conventional WXC-based network, extending the achievable transmission hop count by up to 14 under the examined conditions.

Hybrid FXC-WXC Network with Selective Wavelength Conversion

Figure 1 shows the hybrid FXC-WXC network architecture, in which either FXC or WXC node is selectively deployed in each node. WXC nodes provide wavelength-granular routing, add/drop, grooming, and power equalisation, whereas FXCs provide lower-loss fibre-granular switching. The hybrid FXC-WXC networks form a star-like logical topology, in which WXCs serve as hubs and the transponder banks (TPBs) connected to the FXCs serve as edge nodes. Each optical path traverses at least one WXC, while the number of traversed WXC nodes is limited to suppress transmission impairment.

Under static optical path design, this architecture achieves both high resource-utilisation efficiency and high signal quality. Under dynamic optical path control, however, wavelength resources become unevenly distributed as optical paths are repeatedly established and released, making wavelength contention more likely. As shown in Fig. 2, a conventional WXC-based network can often avoid such contention by assigning the requested optical path to an alternative path. In the hybrid FXC-WXC architecture, by contrast, the coarse-granularity switching of

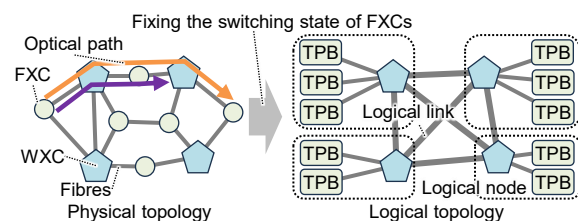


Fig. 1: Logical topology of the hybrid FXC-WXC network.

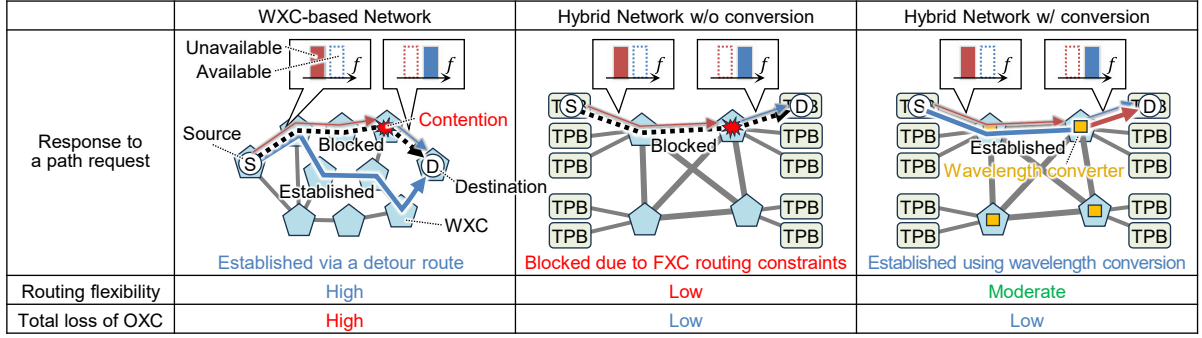


Fig. 2: Challenges of a hybrid FXC-WXC network under a dynamic traffic environment.

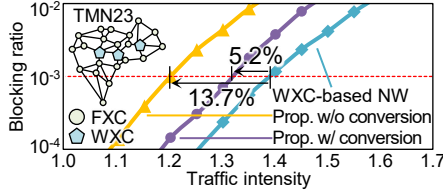


Fig. 3: Blocking characteristics.

FXCs limits routing flexibility, making wavelength contention more likely to result in request blocking. Increasing the number of WXC nodes could mitigate this problem, but would increase node loss, thereby weakening the low-loss advantage of the hybrid FXC-WXC architecture.

To address this issue, we introduce selective wavelength-conversion resources only at WXC nodes. Since wavelength contention arises in wavelength-granular routing, which is performed only at WXC nodes, FXCs do not need wavelength-conversion capability. This selective deployment locally relaxes the wavelength continuity constraint while suppressing additional cost and operational complexity. Furthermore, when wavelength conversion is accompanied by signal regeneration, signal quality can also be restored, extending the feasible transmission reach and improving the applicability of pluggable transceivers in metro and core optical networks.

Numerical Simulations

Numerical simulations were conducted to evaluate the effectiveness of the proposed hybrid FXC-WXC network with selective wavelength conversion. The TMN23 topology was used as an example [23]. The available spectrum was assumed to be 4.8 THz in the extended C-band. Optical path requests were generated under a uniform traffic distribution, and each request carried 400 Gbps on a 100 GHz grid. The traffic intensity was defined as the average number of optical paths per node pair. In the OSNR evaluation, the wavelength selective switch (WSS) loss, matrix switch (MS) loss, fibre attenuation, amplifier noise figure, and transmitter OSNR were set to 8.0 dB, 2.0 dB, 0.18 dB/km, 6.0 dB, and 37.0 dB, respectively. The OSNR threshold for 400ZR+ was set to 24.0 dB [24]. A conventional WXC-based network was used as the benchmark.

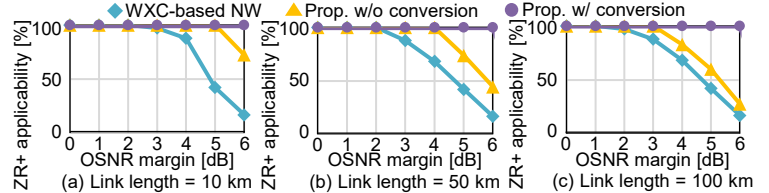


Fig. 4: ZR+ applicability.

In the proposed network, one TPB consisting of 48 transponders was placed at each node, and extra TPBs for wavelength conversion were deployed only at the WXC nodes. Each network was designed to have the same total number of fibres, and dynamic control simulations were performed for three cases: the conventional WXC-based network, the hybrid FXC-WXC network without wavelength conversion, and the hybrid FXC-WXC network with selective wavelength conversion.

Figure 3 shows the blocking ratio as a function of traffic intensity. At a target blocking ratio of 10^{-3} , the hybrid FXC-WXC network without wavelength conversion accommodated 13.7% fewer optical paths than the conventional WXC-based network. Introducing wavelength-converter banks only at the WXC nodes, corresponding to approximately 17% of all transceivers in the network, suppressed this reduction to 5.2%. This result indicates that selective wavelength conversion mitigates the penalty associated with reduced routing flexibility caused by fibre-granular switching at FXC nodes under dynamic traffic conditions.

Figure 4 shows the ZR+ applicability as a function of the OSNR tolerance margin for different link lengths. In the proposed hybrid FXC-WXC architecture, the reduced number of traversed WXC nodes improves signal quality relative to the conventional WXC-based network, thereby enhancing ZR+ applicability. For longer inter-node distances, however, fibre-induced degradation becomes dominant, limiting the gain obtained solely from reducing node loss. In contrast, when signal regeneration is selectively introduced together with wavelength conversion at the WXC nodes, the accumulated degradation can be compensated during transmission, resulting in the highest applicability.

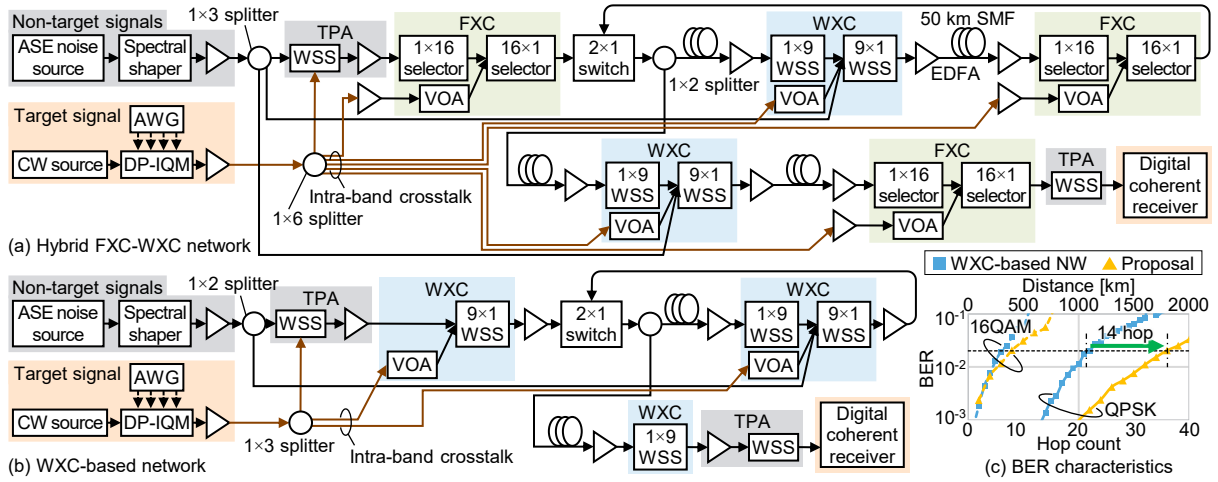


Fig. 5: Transmission experiments: setups for (a) the FXC-WXC network and (b) the WXC-based network; (c) experimental results.

Transmission Experiments

Transmission experiments were conducted to evaluate the transmission performance of the proposed architecture. Figure 5 shows the experimental setups and the corresponding results. Setups (a) and (b) emulate the proposed hybrid FXC-WXC network and a conventional WXC-based network, respectively. In both setups, a 64-Gbaud dual-polarization (DP)-QPSK/16QAM signal was generated using a DP-IQ modulator (IQM) driven by an arbitrary waveform generator (AWG). The generated signal was amplified by an erbium-doped fibre amplifier (EDFA) and split into tributaries. One tributary was used as the target signal, while the others were used to emulate intra-band crosstalk. To emulate non-target signals, 47 wavelength channels with 100-GHz spacing across the extended C-band were generated using an amplified spontaneous emission (ASE) noise source and a spectral shaper. These channels were combined with the target signal by a WSS and then amplified by an EDFA, thereby emulating a transponder aggregator (TPA).

In setup (a), the combined signal was launched into the network through an FXC composed of a 1×16 selector and a 16×1 selector. The signal then propagated through a link-and-node loop consisting of a 2×1 switch, a 1×2 splitter, a 50-km fibre link, a WXC composed of a 1×9 WSS and a 9×1 WSS, another 50-km fibre link, and an FXC. Each fibre link consisted of 50-km single-mode fibre (SMF), with EDFAs placed before and after the fibre span. The EDFA after the FXC was omitted because of the low insertion loss of the FXC. The SMF had a loss coefficient of 0.18 dB/km, a dispersion parameter of 16.25 ps/nm/km, and a nonlinearity coefficient of 1.5 /W/km, while the EDFAs had noise figures of 5 dB. After circulating through the loop, the signal

passed through two additional 50-km fibre links, followed by a WXC and an FXC, respectively. It was then dropped by a WSS emulating a TPA and detected by a digital coherent receiver.

In setup (b), the combined signal was launched into the network through a 9×1 WSS belonging to the WXC. The signal then propagated through a loop consisting of a 2×1 switch, a 1×2 splitter, a 50-km fibre link, and a WXC. After circulating through the loop, the signal passed through an additional 50-km fibre link and was dropped by a 1×9 WSS belonging to the WXC. The received signal was then amplified, passed through a WSS emulating a TPA, and detected by a digital coherent receiver.

Figure 5(c) shows the bit error ratio (BER) as a function of hop count for these cases. Compared with the conventional WXC-only network, the hybrid FXC-WXC network achieves lower BER because the number of traversed WXC filtering stages is reduced and the node loss is smaller. This improvement expands the ZR+ applicability for core/metro-scale optical networks by reducing BER and increasing the achievable transmission reach.

Conclusion

We proposed a low-loss hybrid FXC-WXC network architecture with selective wavelength conversion and evaluated it through numerical simulations and transmission experiments. Numerical results showed that selective wavelength conversion improved accommodation performance and expanded the OSNR margin under dynamic optical path control. Transmission experiments further demonstrated lower BER and better transmission performance than that of a conventional WXC-based network. These results indicate that the proposed architecture is promising for broader deployment of pluggable ZR+ transceivers in metro and core optical networks.

Acknowledgements

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